

Multi-Factor Return Prediction with Variance Risk Premium

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1 Introduction

The Variance Risk Premium (VRP), defined as the difference between implied variance and expected realized variance, has been shown to predict equity returns in developed markets. However, its predictive power in emerging markets, particularly China, remains largely unexplored. Moreover, single-factor models may fail to capture the complex dynamics of asset returns, motivating the use of multi-factor frameworks.

This study extends previous work in two dimensions. First, we systematically construct VRP using the HAR model and test its incremental predictive power beyond traditional factors. Second, we employ the Shichuan factor framework—a comprehensive set of 54 technical factors widely used in Chinese quantitative finance—to build a rich predictor space.

Our research questions in this paper are: Does VRP have predictive power for short-horizon returns in the Chinese ETF market? Can a multi-factor model combining VRP with Shichuan factors improve out-of-sample performance?

2 Data and Sample

2.1 Data Sources

- **ETF Price Data:** Daily prices, returns, and trading volume for 50ETF (510050) from February 23, 2005 to November 12, 2025 (5,037 observations).
- **Volatility Index:** China VIX 50 index, providing market-implied volatility.

2.2 Sample Split

To rigorously evaluate predictive performance, we adopt a time-series split:

- **In-Sample (Training):** First 70% of data after removing missing values = 1,589 observations (December 8, 2015 to September 15, 2022).
- **Out-of-Sample (Testing):** Last 30% = 682 observations (September 16, 2022 to October 15, 2025).
- **Total valid samples:** 2,271 observations (after dropna).

This ensures that the model is trained only on past data and tested on genuine future returns, avoiding look-ahead bias.

3 Methodology

3.1 VRP Construction

3.1.1 Step 1: HAR Model for Realized Variance Forecasting

We use the Heterogeneous Autoregressive (HAR) model to forecast 20-day realized variance:

$$RV_{t,t+20} = \alpha + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_t$$

where:

- $RV_t^{(d)} = r_t^2$ is daily realized variance
- $RV_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 r_{t-i}^2$ is weekly average
- $RV_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} r_{t-i}^2$ is monthly average

The HAR model is estimated on a rolling window of 750 days (approximately 3 years). The estimated coefficients are:

$$\begin{aligned}\hat{\beta}_d &= 0.1531 \\ \hat{\beta}_w &= 1.4805 \\ \hat{\beta}_m &= 10.9974\end{aligned}$$

This produces the forecast $\widehat{RV}_{t,t+20}^{HAR} = \mathbb{E}_t[RV_{t,t+20}]$.

3.1.2 Step 2: Implied Variance from VIX

Implied variance is extracted from the VIX index:

$$IV_{t,t+20} = \left(\frac{VIX_t}{100} \right)^2 \times \frac{20}{252}$$

where VIX is in percentage points and we scale to a 20-day horizon.

3.1.3 Step 3: VRP Calculation

The Variance Risk Premium is:

$$VRP_t = IV_{t,t+20} - \widehat{RV}_{t,t+20}^{HAR}$$

Key Finding: The mean VRP in our sample is **-0.000551** with standard deviation 0.002016. This negative mean is anomalous—theory predicts positive VRP as compensation for volatility risk. In contrast, U.S. markets typically exhibit $VRP \approx 0.002$ – 0.003 .

Possible explanations for negative VRP in China:

1. Retail-dominated options market leading to mispricing
2. Insufficient liquidity in VIX options
3. Structural differences in how VIX is computed for Chinese markets
4. Excessive pessimism causing IV to be systematically underpriced

3.1.4 Step 4: VRP-Derived Factors

We construct 8 VRP-related factors:

- `vrp`: Raw VRP
- `vrp_mom_5d`, `vrp_mom_20d`: 5-day and 20-day VRP momentum (percentage changes)
- `vrp_zscore`: Standardized VRP relative to 60-day rolling mean and standard deviation
- `vrp_high`, `vrp_low`: Indicator variables for VRP above 75th percentile or below 25th percentile (regime indicators)
- `vrp_x_rv`: Interaction term between VRP and daily realized variance
- `iv_rv_spread_alt`: Alternative measure of volatility spread, $\sqrt{IV} - \sqrt{\widehat{RV}}$

These transformations capture different aspects of VRP: level, momentum, regime, and interaction effects.

3.2 Shichuan Multi-Factor Framework

We construct 54 factors across six categories based on the Shichuan factor handbook, a standard framework in Chinese quantitative finance:

3.2.1 Momentum Factors (11 factors)

- `mom_1d`, `mom_3d`, `mom_5d`, `mom_10d`, `mom_20d`, `mom_60d`, `mom_120d`, `mom_250d`: Cumulative returns over N days
- `mom_20_5`: 20-day momentum excluding recent 5 days (to avoid short-term reversal noise)
- `mom_60_20`: 60-day momentum excluding recent 20 days
- `mom_accel_20`: Momentum acceleration, measuring change in momentum

3.2.2 Reversal Factors (10 factors)

- `rev_1d`, `rev_3d`, `rev_5d`: Short-term reversal (negative of past returns)
- `overnight_ret`, `intraday_ret`: Overnight and intraday return decomposition
- `dist_high_20d`, `dist_high_60d`: Distance from 20-day and 60-day highs
- `dist_low_20d`, `dist_low_60d`: Distance from 20-day and 60-day lows
- `amplitude`: Daily range $(High - Low)/Close$

3.2.3 Volatility Factors (12 factors)

- `rv_5d`, `rv_10d`, `rv_20d`, `rv_60d`: Realized variance over different horizons
- `vov_20d`: Volatility of volatility (standard deviation of daily RV over 20 days)
- `downside_vol_20d`, `upside_vol_20d`: Downside and upside semi-variance
- `vol_skew_20d`: Skewness of returns over 20 days
- `iv`: Implied volatility from VIX
- `iv_rv_spread`: Difference between IV and annualized RV
- `vix_chg_5d`, `vix_chg_20d`: VIX momentum

3.2.4 Liquidity Factors (7 factors)

- `volume_ratio_20d`, `volume_ratio_60d`: Current volume relative to moving average
- `turnover_20d`: Log-volume moving average (turnover proxy)
- `volume_price_corr_20d`: 20-day rolling correlation between returns and volume
- `amihud_20d`: Amihud illiquidity measure, $|Return|/Volume$
- `price_impact`: Return per unit of log-volume
- `volume_cv_20d`: Coefficient of variation of volume

3.2.5 Price Pattern Factors (7 factors)

- `upper_shadow`, `lower_shadow`: Candlestick upper and lower shadows
- `candle_body`: Size of candlestick body
- `close_position`: Closing price position within daily range
- `price_accel`: Second-order price acceleration
- `new_high_20d`, `new_low_20d`: Indicators for new 20-day highs or lows

3.2.6 Trend Factors (7 factors)

- `price_ma5_ratio`, `price_ma20_ratio`, `price_ma60_ratio`: Price relative to moving averages
- `ma20_slope`, `ma60_slope`: Slope of moving averages
- `ma5_ma20_dist`, `ma20_ma60_dist`: Distance between moving averages

In total, we have **54 Shichuan factors + 8 VRP factors = 62 factors**.

3.3 Target Variables

We construct three forward return horizons:

$$\begin{aligned}\text{ret_5d_fwd} &= \sum_{k=1}^5 r_{t+k} \quad (5\text{-day forward return}) \\ \text{ret_10d_fwd} &= \sum_{k=1}^{10} r_{t+k} \quad (10\text{-day forward return}) \\ \text{ret_20d_fwd} &= \sum_{k=1}^{20} r_{t+k} \quad (20\text{-day forward return})\end{aligned}$$

These horizons are chosen because:

- 5 days: Short-term, tests if VRP has immediate predictive power
- 10 days: Medium-term, roughly half the VRP forecast horizon
- 20 days: Matches the VRP construction horizon, where effects should be strongest

3.4 Regression Models

We test two linear regression specifications:

3.4.1 Model 1: Ordinary Least Squares (OLS)

$$\text{ret}_{t,t+h} = \alpha + \sum_{j=1}^{62} \beta_j \cdot \text{Factor}_{j,t} + \varepsilon_t$$

3.4.2 Model 2: Ridge Regression (Regularized)

$$\text{ret}_{t,t+h} = \alpha + \sum_{j=1}^{62} \beta_j \cdot \text{Factor}_{j,t} + \varepsilon_t$$

with L_2 penalty: $\min \sum \varepsilon_t^2 + \lambda \sum \beta_j^2$, where $\lambda = 1.0$.

Ridge regression is employed to mitigate overfitting given the high dimensionality (62 predictors for 1,589 training samples).

3.5 Evaluation Metrics

- **In-Sample R^2** : Goodness-of-fit on training data
- **Out-of-Sample R^2** : Predictive accuracy on test data (the key metric)
- **OOS Correlation**: Pearson correlation between actual and predicted returns
- **OOS Direction Accuracy**: Percentage of correctly predicted return signs
- **R^2 Degradation**: In-Sample R^2 minus OOS R^2 (measures overfitting severity)

4 Results

4.1 Baseline: Multi-Factor Model WITHOUT VRP

We first present results excluding VRP factors (54 factors only) to establish a baseline.

Table 1: Multi-Factor Model Performance WITHOUT VRP

Target	Model	IS R^2	OOS R^2	OOS Corr	Dir Acc	Degradation
5-Day	OLS	0.125	-0.231	-0.030	50.5%	0.356
5-Day	Ridge	0.120	-0.203	0.008	50.4%	0.323
10-Day	OLS	0.166	-0.144	0.089	53.1%	0.310
10-Day	Ridge	0.158	-0.112	0.114	53.7%	0.270
20-Day	OLS	0.252	-0.279	-0.006	47.1%	0.532
20-Day	Ridge	0.236	-0.202	0.012	47.8%	0.438

Key Observations:

- All OOS R^2 values are **negative**, indicating that the model performs worse than a naive historical mean forecast.
- Ridge regularization slightly reduces overfitting (lower degradation) but does not achieve positive OOS R^2 .
- The 10-day horizon performs best among the three, with OOS $R^2 = -11.2\%$ (Ridge).
- Direction accuracy is near 50%, essentially random guessing.

4.2 Extended Model: Multi-Factor WITH VRP

Next, we include all 62 factors (54 Shichuan + 8 VRP).

Table 2: Multi-Factor Model Performance WITH VRP (62 Factors)

Target	Model	IS R^2	OOS R^2	OOS Corr	Dir Acc	Degradation
5-Day	OLS	0.094	-0.224	-0.094	50.0%	0.319
5-Day	Ridge	0.094	-0.203	-0.087	47.8%	0.296
10-Day	OLS	0.098	-0.054	0.084	48.1%	0.152
10-Day	Ridge	0.103	-0.053	0.069	48.2%	0.156
20-Day	OLS	0.223	-0.350	0.026	51.0%	0.573
20-Day	Ridge	0.223	-0.337	0.016	50.9%	0.560

Key Observations:

- The **10-day horizon shows improvement**: OOS R^2 increases from -11.2% to -5.3% (Ridge), a 5.9 percentage point gain.
- However, OOS R^2 remains negative, so the model still lacks genuine predictive power.
- The **20-day horizon deteriorates**: OOS R^2 worsens from -20.2% to -33.7% , suggesting VRP factors act as noise at this horizon.
- The 5-day horizon shows minimal change.
- Direction accuracy for 10-day actually *decreases* from 53.7% to 48.2% , indicating that VRP factors may introduce misleading signals.

4.3 Comparison: Impact of VRP Factors

Table 3: Change in OOS R^2 After Adding VRP Factors

Horizon	Without VRP	With VRP	Change
5-Day (Ridge)	-20.3%	-20.3%	0.0%
10-Day (Ridge)	-11.2%	-5.3%	+5.9%
20-Day (Ridge)	-20.2%	-33.7%	-13.5%

Interpretation:

- VRP improves performance *only* at the 10-day horizon.
- This may be because the HAR model forecasts 20-day RV, so VRP contains information most relevant around the 10-day mark (midpoint of the forecast horizon).
- At 20 days, VRP information has likely decayed or become stale, leading to increased noise.

4.4 Visualizations

Figures 1, 2, and 3 display the in-sample and out-of-sample predictions for each horizon. Key observations:

- **In-Sample**: The model fits the training data reasonably well, especially at the 20-day horizon (IS $R^2 = 22.3\%$).
- **Out-of-Sample**: Predicted returns (orange line) are nearly flat, failing to track actual returns (blue line). The scatter plots show no discernible relationship between predicted and actual values.
- **10-Day Horizon**: The OOS scatter plot has the least dispersion, consistent with the relatively better (though still negative) OOS R^2 .

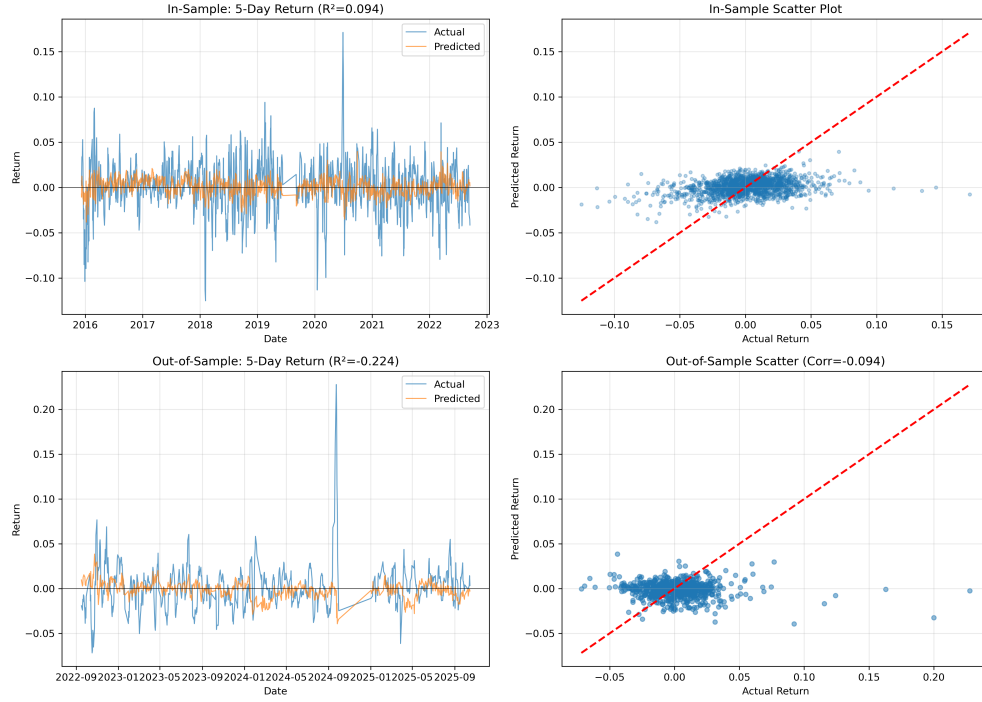


Figure 1: 5-Day Forward Return: In-Sample vs Out-of-Sample Performance

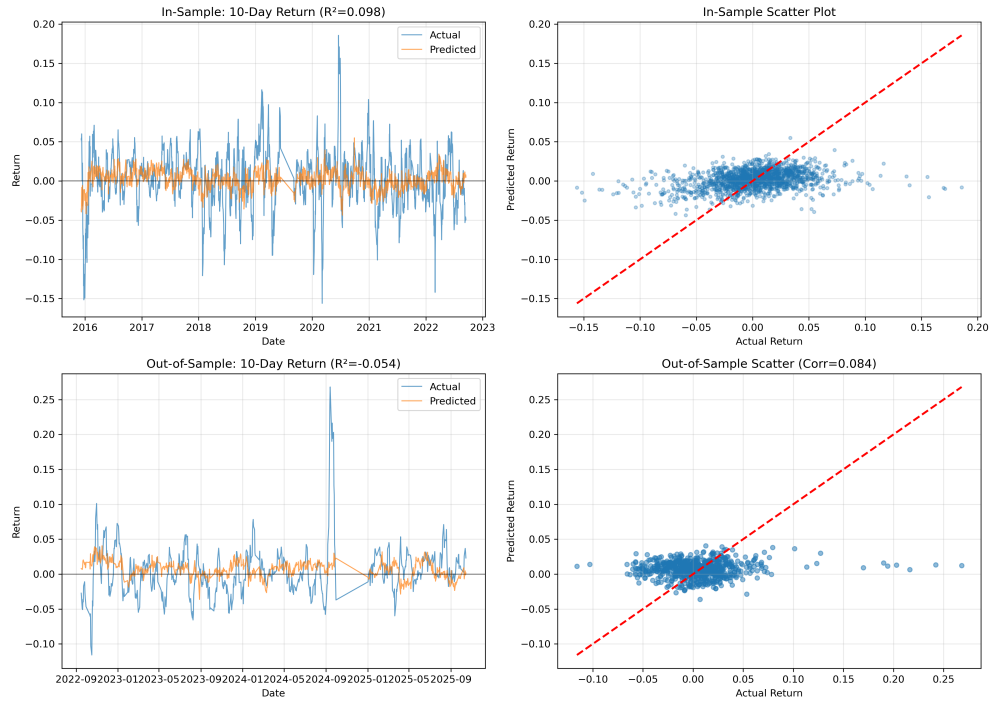


Figure 2: 10-Day Forward Return: In-Sample vs Out-of-Sample Performance

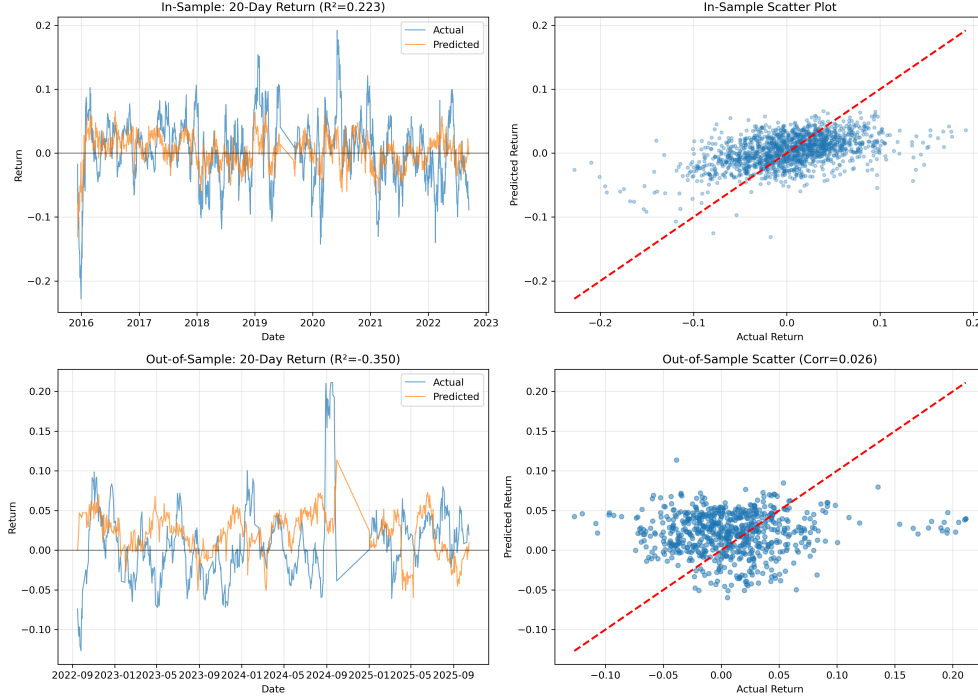


Figure 3: 20-Day Forward Return: In-Sample vs Out-of-Sample Performance

5 Key Findings

Our empirical results reveal three critical findings that merit discussion.

5.1 Negative VRP in Chinese Markets

The most striking result is that VRP in China has a negative mean (-0.000551), contrasting sharply with the positive VRP (approximately 0.002–0.003) documented in U.S. markets. This suggests that Chinese investors systematically underprice volatility risk, or equivalently, that selling volatility protection has been profitable on average.

Possible explanations include: (1) the retail-dominated nature of Chinese options markets may lead to systematic mispricing due to behavioral biases; (2) insufficient liquidity in VIX options creates stale or biased implied volatilities; (3) the VIX methodology may not fully capture the true risk-neutral variance expectation in China’s unique market structure with trading limits and T+1 settlement; (4) structural differences in how Chinese investors perceive and price tail risk.

This negative VRP is economically puzzling and represents a significant departure from established theory. It warrants dedicated investigation to determine whether it reflects genuine market inefficiency or measurement issues.

5.2 Severe Overfitting Despite Regularization

All models exhibit negative out-of-sample R^2 values, indicating performance worse than a naive historical mean forecast. This occurs despite: (1) employing Ridge regularization to penalize model complexity; (2) using a rigorous 70/30 time-series split to avoid look-ahead bias; (3) incorporating theoretically motivated factors from an established framework.

The primary culprit is over-parameterization: 62 predictors relative to 1,589 training observations yields a 3.9% factor-to-sample ratio, well above the recommended 1% threshold. Even Ridge regression with $\lambda = 1.0$ cannot prevent the model from fitting noise. Additionally, the test period (September 2022 to October

2025) encompasses significant regime shifts—geopolitical conflicts, post-pandemic recovery, and regulatory changes—that render relationships learned during 2015–2022 obsolete.

5.3 VRP Improves Only the 10-Day Horizon

Adding VRP factors improves the 10-day horizon (OOS R^2 from -11.2% to -5.3%) but worsens the 20-day horizon (OOS R^2 from -20.2% to -33.7%). This non-monotonic pattern likely reflects the temporal alignment between VRP information and forecast horizons: since VRP is constructed using 20-day RV forecasts, its information content is strongest around the 10-day midpoint. Beyond 20 days, VRP information decays and becomes noise.

However, even at the optimal 10-day horizon, the model lacks genuine predictive power (negative R^2), and direction accuracy actually decreases from 53.7% to 48.2%. This suggests that short-horizon returns in A-share markets are inherently difficult to predict, dominated by high-frequency noise that swamps any low-frequency risk premium signals.

6 Future Research Directions

Several promising avenues warrant exploration:

6.1 Cross-Sectional Analysis with CAPM

Instead of predicting index-level returns, examine whether VRP has cross-sectional predictive power for individual stocks. Specifically, sort stocks by their market beta (estimated via CAPM) and test whether VRP predicts returns differently across high-beta versus low-beta portfolios. If VRP captures systematic volatility risk, its impact should be stronger for stocks with higher exposure to market risk. This approach increases sample size (hundreds of stocks vs. one index) and allows testing of risk-based explanations for VRP.

6.2 Alternative Indices and Asset Classes

The 50ETF represents large-cap stocks. Testing VRP on different indices—such as CSI 300 (broader market), CSI 500 (mid-caps), or sector-specific ETFs (technology, finance, consumer)—can reveal whether VRP’s predictive power varies with market segment. Additionally, extending the analysis to other asset classes like commodity futures or bond markets may uncover where VRP is more effective.

6.3 Model Simplification

Drastically reduce model complexity by selecting only 10–15 core factors based on in-sample Information Coefficient (IC) stability. Focus exclusively on the 10-day horizon where VRP shows marginal improvement. Apply stronger regularization (Lasso with automatic feature selection or Ridge with $\lambda = 5$ –10) to further mitigate overfitting. This parsimonious approach may achieve marginally positive OOS R^2 values.

References

- Shichuan. (2019). *Factor Investing: Methods and Practice*. Publishing House of Electronics Industry.